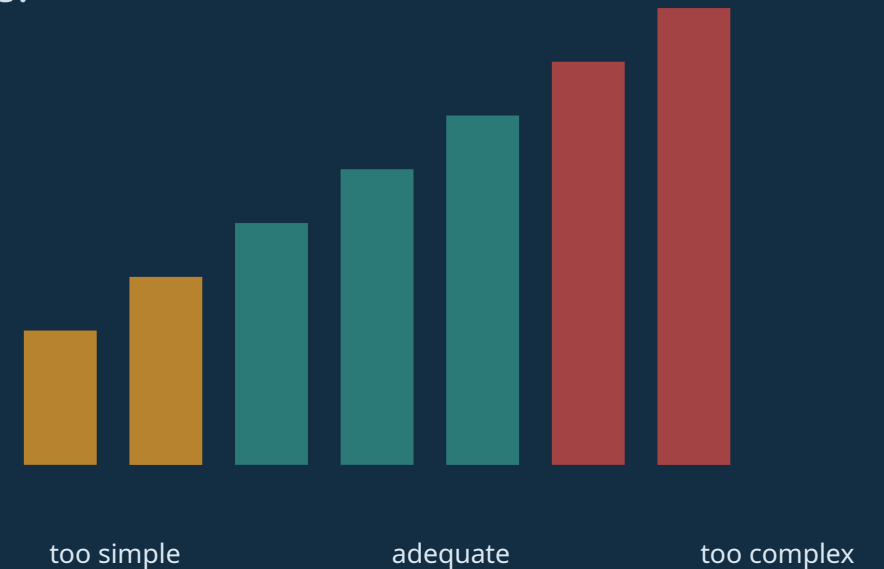


Departures from the Mean Model: Underfitting and Overfitting

What happens when the fitted model is too small—or unnecessarily large?

$$Y = X\beta + \varepsilon$$



Learning objectives

Treat model complexity as a linear-model problem: subspaces, bias, variance, and inference.

ROADMAP

- Distinguish mean-model misspecification from violations of the error assumptions.
- Describe underfitting as failure of the fitted column space to contain the true mean vector.
- Derive the bias introduced by omitted structure and its effect on residual sums of squares.
- Explain why overfitting reduces training SSE but increases estimation variance and consumes residual degrees of freedom.
- Use nested-model geometry to compare underfit, adequate, and overfit specifications.
- Connect model complexity to prediction error, diagnostics, hypothesis tests, and model-selection procedures.

Where this fits in linear-model theory

The central assumption is about the mean vector—not merely about straight lines.

BIG PICTURE

$$Y = \mu + \varepsilon, \quad E(\varepsilon) = 0, \quad \text{Var}(\varepsilon) = \sigma^2 I$$

linear model assumption: $\mu \in \mathcal{C}(X)$

Underfitting

The chosen model space is too small: $\mu \notin \mathcal{C}(X)$. Some systematic mean structure is forced into the residuals.

Overfitting

The chosen model space is larger than needed. It may still contain μ , so the model can be correct but inefficient.

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Mean-model misspecification

Underfitting as projection of the true mean onto the wrong subspace.

A useful distinction: wrong mean model vs. wrong error model

ASSUMPTIONS

Issue	Mathematical statement	Typical consequence
Underfit mean	$\mu \notin C(X)$	bias / structured residuals
Heteroskedasticity	$\text{Var}(\varepsilon) \neq \sigma^2 I$	wrong standard errors
Correlation	$\text{Cov}(\varepsilon_i, \varepsilon_j) \neq 0$	inefficient OLS / wrong SEs
Non-normality	ε not Gaussian	finite-sample t/F theory changes
Overfit mean	$C(X)$ larger than needed	higher variance / lower df

This lecture focuses on the size and content of the mean space $C(X)$.

The projection target when the model is wrong

GEOMETRY

OLS always projects Y onto $C(X)$, whether or not the true mean lies there.

$$\hat{Y} = HY, \quad H = X(X^T X)^{-1} X^T$$

$$E(\hat{Y}) = H\mu$$

If $\mu \in C(X)$

$H\mu = \mu$. The fitted mean is unbiased for the true mean vector.

If $\mu \notin C(X)$

$H\mu$ is only the closest point in $C(X)$ to μ . The missing component $(I-H)\mu$ is systematic bias.

Underfitting: the bias vector

UNDERFITTING

$$b_{\mu} = E(\hat{Y}) - \mu = H\mu - \mu = -(I - H)\mu$$

The vector $(I - H)\mu$ is the part of the true mean that the chosen model cannot represent.

Geometric interpretation

Underfitting is not vague “simplicity.” It is a precise subspace statement: the true mean has a component orthogonal to the fitted model space.

Statistical interpretation

That missing component behaves like deterministic structure left in the residuals. Repeated samples reproduce the same residual pattern on average.

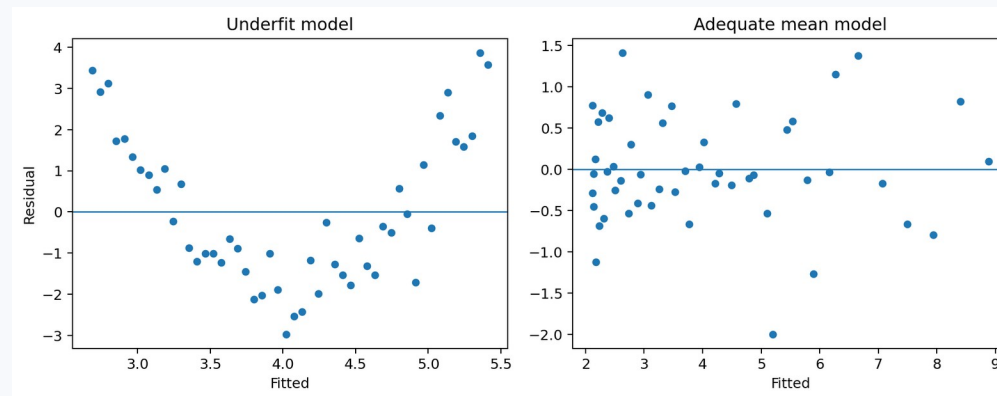
Expected residuals under misspecification

RESIDUALS

$$e = (I - H)Y$$

$$E(e) = (I - H)\mu$$

Under a correctly specified mean model, $E(e)=0$. Under underfitting, the residual vector has a nonzero mean.



Expected SSE under an underfit model

QUADRATIC FORMS

Let $M = I - H$. Then $SSE = Y^T M Y$.

$$E(Y^T M Y) = \text{tr}(M \sigma^2 I) + \mu^T M \mu$$

$$E(SSE) = (n - p) \sigma^2 + \|(I - H) \mu\|^2$$

Noise term

$(n - p) \sigma^2$ is the expected residual variation when the mean model is correct.

Lack-of-fit term

$\|(I - H) \mu\|^2$ is systematic variation that the fitted mean space cannot explain.

MSE is no longer unbiased for σ^2

CONSEQUENCE

$$MSE = \frac{SSE}{n - p}$$

$$E(MSE) = \sigma^2 + \frac{\|(I - H)\mu\|^2}{n - p}$$

Key consequence

When the model is underfit, residual mean square mixes random noise with lack of fit. It generally overestimates σ^2 .

Why inference suffers

Standard errors based on MSE are no longer estimating the intended noise variance, and coefficient tests can answer the wrong scientific question.

Omitted-variable bias in coefficient form

OMITTED STRUCTURE

Suppose the true model contains both X and omitted covariates Z :

$$Y = X\beta + Z\gamma + \varepsilon$$

but we fit only X . Then

$$E(\hat{\beta}_X) = \beta + (X^T X)^{-1} X^T Z\gamma$$

No bias if...

$X^T Z\gamma = 0$. The omitted signal is orthogonal to the included design columns.

Bias if...

The omitted signal is correlated with directions already represented by X . The included coefficients absorb part of the omitted effect.

Underfitting includes more than omitted covariates

EXAMPLES

Nonlinearity

Fit $\beta_0 + \beta_1 x$ when $E(Y | x)$ is curved.

Missing interaction

Fit additive effects when the effect of x depends on z .

Wrong functional form

Use x instead of $\log(x)$, spline terms, thresholds, etc.

Missing factor levels / structure

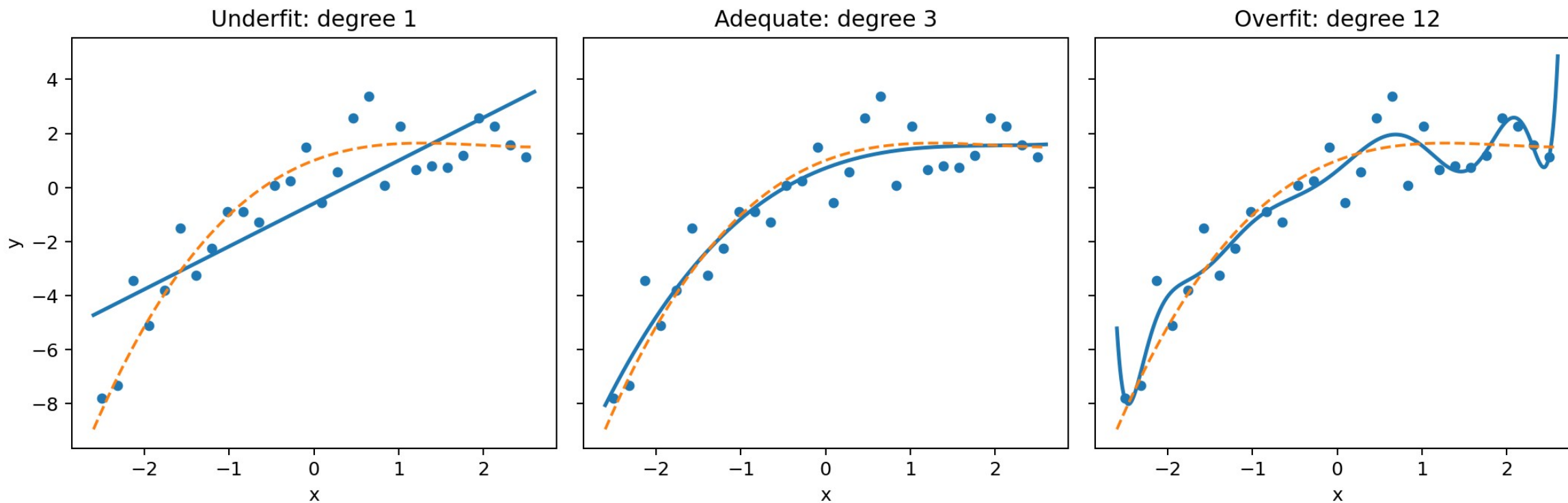
Collapse scientifically distinct groups into a common mean.

Overly restrictive constraints

Impose parameter restrictions not satisfied by the true mean.

A simple picture: too little, enough, too much

POLYNOMIAL EXAMPLE



Dashed curve = true mean function. Solid curve = fitted model.

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Overfitting

When the fitted space is larger than the true mean space.

Overfitting is different from misspecification

IMPORTANT DISTINCTION

$$\mathcal{C}(X_{\text{true}}) \subsetneq \mathcal{C}(X_{\text{fit}})$$

Mean model can still be correct

If the true mean lies in the larger fitted space, then $H\mu = \mu$. OLS fitted values remain unbiased for μ .

But estimation is less efficient

Extra directions are estimated from noise. Coefficient variances increase, residual df decrease, and prediction can become less stable.

Overfitting is usually an efficiency and generalization problem—not necessarily a violation of $E(Y)=X\beta$.

Training SSE cannot increase when predictors are added

NESTED SPACES

For nested spaces $C(X_0) \subset C(X_1)$,

$$SSE_1 \leq SSE_0$$

because the larger model minimizes squared error over a superset of candidate fitted values.

$$SSE_0 - SSE_1 = Y^T (H_1 - H_0) Y \geq 0$$

Crucial warning

A lower training SSE is therefore not evidence that the larger model is closer to the true mean.

Why extra predictors increase coefficient variance

VARIANCE

$$\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$$

Adding a predictor can make the design directions less distinct, especially when the new predictor is correlated with existing columns.

Geometric view

The coefficient vector is a coordinate representation of the fitted projection. Nearly redundant columns make those coordinates unstable.

Inferential view

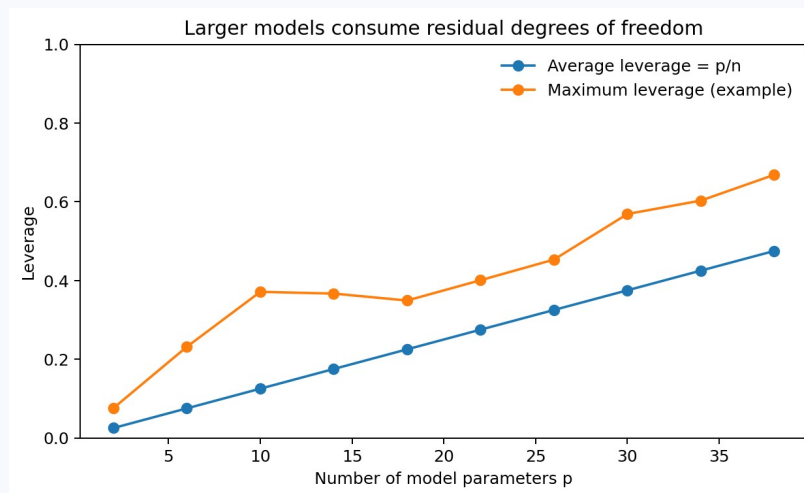
Standard errors grow, confidence intervals widen, and individual t statistics lose power—even though SSE weakly decreases.

Residual degrees of freedom are a finite resource

DEGREES OF FREEDOM

$$df_E = n - p$$

Every additional linearly independent model direction consumes one residual degree of freedom.



Average leverage satisfies $\text{tr}(H)/n = p/n$.

What happens as p approaches n ?

NEAR SATURATION

Regime	Geometry	Consequence
$p \ll n$	large residual subspace	stable σ^2 estimation
p moderately large	smaller residual subspace	more leverage / variance
$p = n$, full rank	$H = I$	$SSE = 0$; no residual df
$p > n$	X cannot have full column rank	β not uniquely identified

Perfect interpolation at $p=n$ says nothing about out-of-sample accuracy.

Overfitting and leverage

HAT MATRIX

$$h_{ii} = \mathbf{H}_{ii}, \quad \sum_{i=1}^n h_{ii} = p$$

Average leverage

p/n . Larger models necessarily raise the average diagonal element of $\mathbf{H}$.

Practical effect

Some observations can exert stronger control over their own fitted values. Residuals shrink mechanically at high leverage points.

$$\text{Var}(e_i) = \sigma^2(1 - h_{ii})$$

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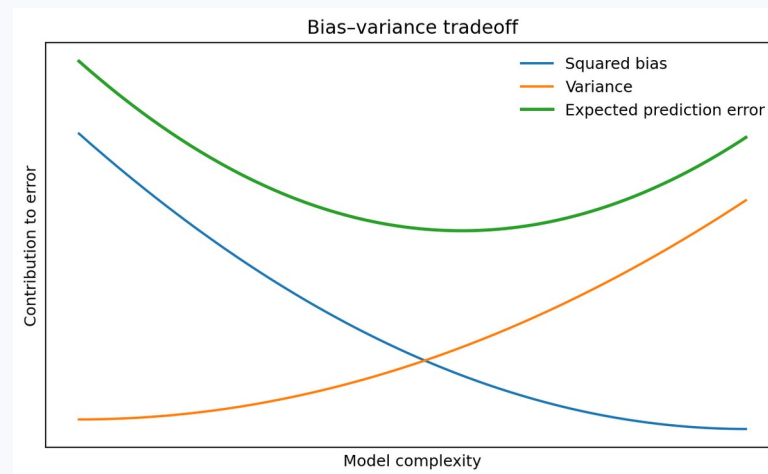
Bias–variance tradeoff

Why the “best” model is not generally the model with the smallest training error.

Prediction error decomposes into bias and variance

PREDICTION

For a target point x_0 , write the prediction error schematically as



Underfitting and overfitting occupy opposite sides of the tradeoff

TRADEOFF

Underfitting

High squared bias, usually lower variance. The model systematically misses real structure.

Overfitting

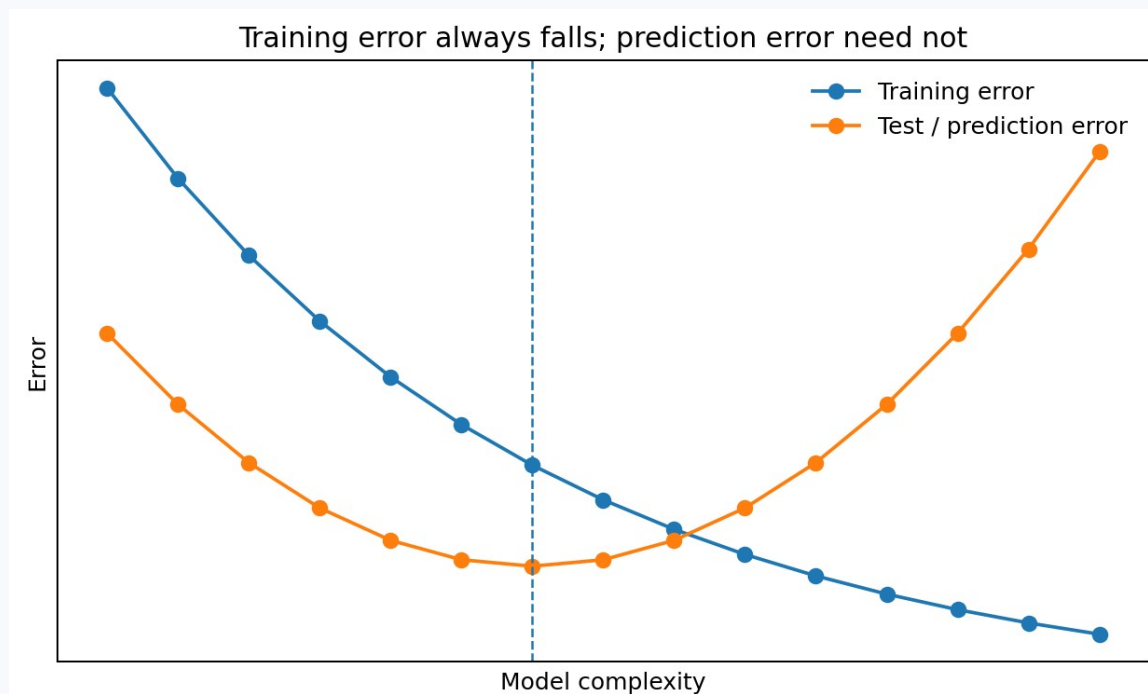
Low training bias, potentially high variance. The model learns directions that are mostly noise.

$$\text{good model complexity} \neq \arg \min_p SSE_p$$

The inferential target and intended use—estimation, testing, or prediction—matter.

Training error versus prediction error

GENERALIZATION



Training criterion

SSE always weakly decreases as a nested model grows.

Prediction criterion

Out-of-sample error may be U-shaped because variance eventually dominates the reduction in bias.

Adjusted R^2 tries to penalize gratuitous complexity

MODEL COMPARISON

$$R^2 = 1 - \frac{SSE}{SST} \qquad R^2_{adj} = 1 - \frac{SSE/(n - p)}{SST/(n - 1)}$$

Ordinary R^2

Never decreases when regressors are added. It rewards training fit only.

Adjusted R^2

Penalizes the loss of residual degrees of freedom. It can decrease when a new term does not earn its complexity cost.

Extra sum of squares separates evidence from automatic fit improvement

NESTED TEST

Suppose X_0 is the smaller model and X_1 adds q directions.

$$SS_{extra} = SSE_0 - SSE_1 = Y^T (H_1 - H_0) Y$$

$$F = \frac{(SSE_0 - SSE_1) / q}{SSE_1 / (n - p_1)}$$

If added directions are unnecessary

The improvement is attributable to fitting noise. Under H_0 , the numerator behaves like a χ^2_q contribution.

If added directions contain signal

The extra quadratic form becomes noncentral; the reduction in SSE is systematically larger.

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Consequences for inference

How model size changes coefficient meaning, standard errors, tests, and residual distributions.

Underfitting can invalidate familiar inferential interpretations

INFERENCE

- Coefficient estimators target the best projection of μ onto $C(X)$, not necessarily the scientific parameters from the omitted larger model.
- Residuals have nonzero mean: $E(e) = (I - H)\mu$.
- SSE/σ^2 is generally noncentral chi-squared rather than central chi-squared.
- MSE is inflated by lack of fit and no longer estimates σ^2 alone.
- t and F procedures may reject because of omitted structure, not because the tested effect itself is scientifically meaningful.

The residual SSE becomes noncentral under mean misspecification

QUADRATIC FORM

With $M=I-H$ and $Y \sim N(\mu, \sigma^2 I)$,

$$\frac{Y^T M Y}{\sigma^2} \sim \chi_{n-p}^2(\lambda), \quad \lambda = \frac{\mu^T M \mu}{\sigma^2} = \frac{\|(I - H)\mu\|^2}{\sigma^2}$$

Correct model

$M\mu=0 \Rightarrow \lambda=0$. The residual sum of squares has the central $\chi^2_{\{n-p\}}$ distribution used in classical theory.

Underfit model

$M\mu \neq 0 \Rightarrow \lambda > 0$. Systematic lack of fit shifts the residual distribution upward.

Overfitting: classical inference can still be exact—conditionally

IMPORTANT NUANCE

If the larger fixed model contains the true mean and is chosen independently of Y , then the usual normal linear-model theory still applies.

Still valid

Unbiased fitted mean, central residual χ^2 law, exact t/F tests for prespecified hypotheses, and correct nominal confidence coverage.

But less efficient

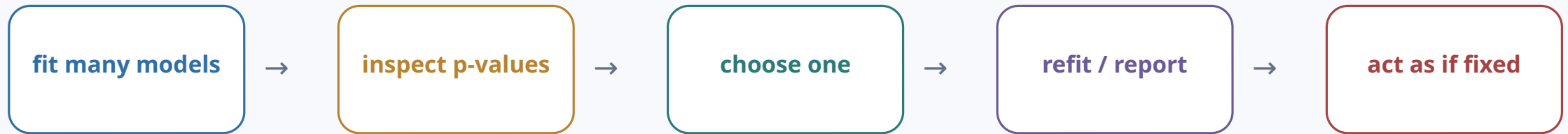
Larger coefficient variances, fewer residual degrees of freedom, wider intervals, and lower power for many contrasts.

Overfitting becomes especially problematic when the same data are used to search among many models and then classical inference is reported as if the model had been fixed in advance.

Data-driven model selection changes the inferential problem

SELECTION

A common workflow is



Why this matters

The selected model is random. Naive standard errors, p-values, and confidence intervals ignore the selection event and can be anti-conservative.

Conceptual distinction

“A prespecified large model” and “a model chosen because it looked best in these data” are not the same inferential object.

05

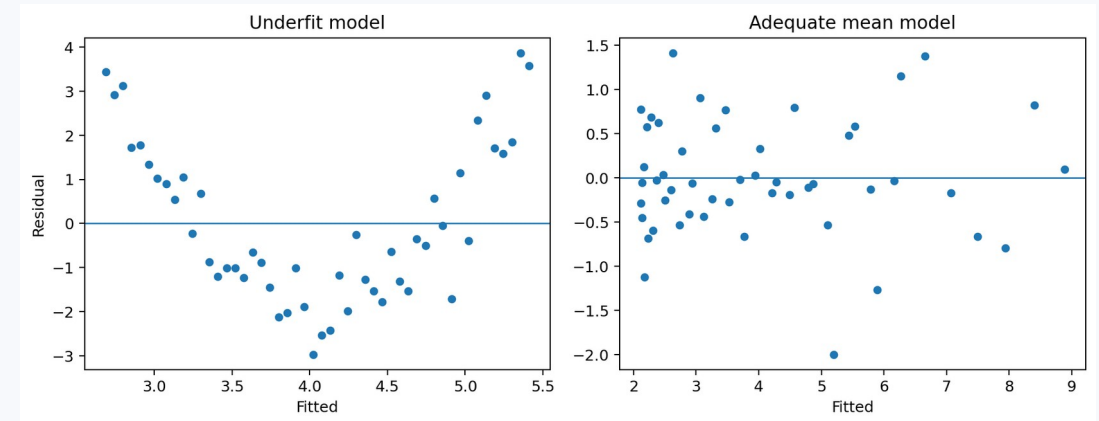
Diagnosing and controlling complexity

Residual structure, lack-of-fit tests, validation, and principled model comparison.

Residual diagnostics can reveal underfitting

DIAGNOSTICS

- Curvature in residual-versus-fitted or residual-versus-predictor plots → missing nonlinear structure.
- Group-specific patterns → omitted factor effects or interactions.
- Systematic runs in ordered residuals → missing time or spatial structure.
- Residual structure should be interpreted against the design: random scatter is expected only after the mean model is adequate.



Caution

Residual diagnostics also respond to heteroskedasticity, correlation, and outliers. A pattern does not uniquely identify underfitting.

Replicated x values permit a formal lack-of-fit decomposition

LACK OF FIT

When several observations share the same predictor settings, residual SSE can be split into

$$SSE = SS_{PE} + SS_{LOF}$$

Pure error

Variation among replicate observations at identical predictor settings. This estimates σ^2 without relying on the mean model.

Lack of fit

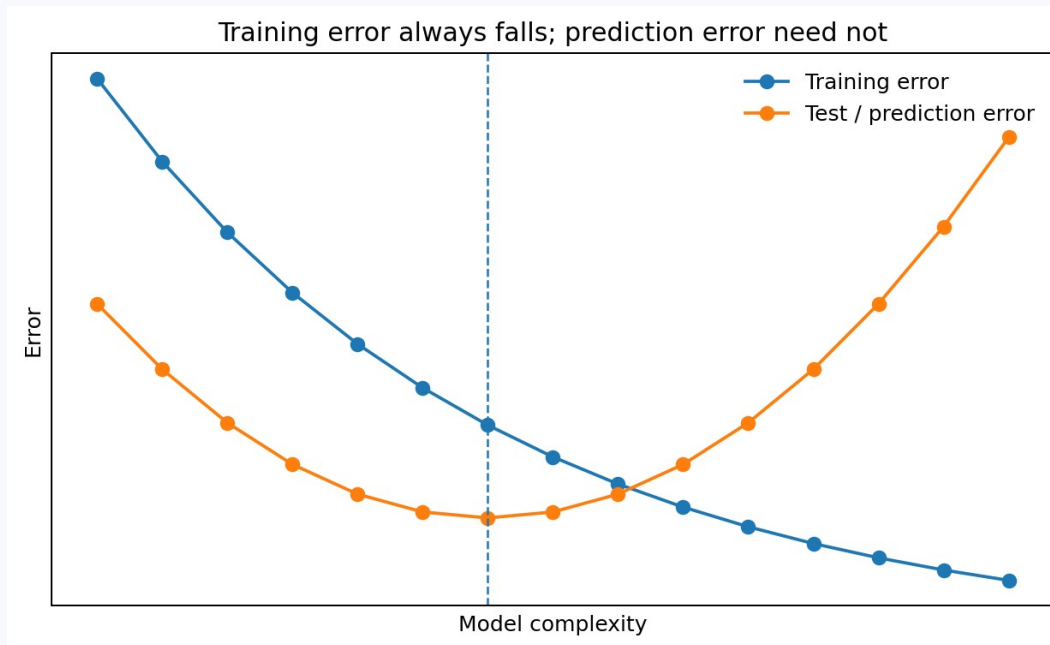
Additional residual variation caused by the fitted mean function failing to reproduce the replicate means.

$$F_{LOF} = \frac{MS_{LOF}}{MS_{PE}}$$

Cross-validation targets prediction rather than training fit

VALIDATION

The idea: evaluate each candidate model on observations that were not used to estimate its coefficients.



Why useful

Validation error can rise when a model starts fitting idiosyncratic noise, even though training SSE continues to fall.

What it does not solve

Cross-validation chooses a prediction procedure. It does not automatically restore classical post-selection p-values or confidence intervals.

AIC and BIC encode explicit fit-complexity tradeoffs

INFORMATION CRITERIA

$$AIC = -2\ell(\hat{\theta}) + 2k \quad BIC = -2\ell(\hat{\theta}) + k \log n$$

AIC intuition

Penalizes complexity with roughly 2 units per estimated parameter; often motivated by expected predictive discrepancy.

BIC intuition

Penalty increases with $\log n$; tends to favor smaller models as sample size grows under its model-selection framework.

Both recognize the same principle: a reduction in SSE must be large enough to justify the added model dimension.

A theory-first workflow for choosing model complexity

WORKFLOW

1

Specify plausible mean structure from science/design before looking at significance.

2

Fit a model rich enough to represent known nonlinearities, interactions, and design features.

3

Inspect residuals for systematic mean structure that remains.

4

Use nested tests when the scientific question is a prespecified restriction.

5

Use validation or information criteria when prediction / model selection is the goal.

6

Distinguish confirmatory inference from exploratory model search.

Common misconceptions

PITFALLS

“More predictors always improve the model.”

They improve training fit, not necessarily the approximation to μ or predictive performance.

“Overfitting means the linear-model assumptions are false.”

Not necessarily. A larger fixed model can contain the true mean and satisfy all classical assumptions.

“High R^2 proves the model is adequate.”

R^2 cannot diagnose misspecified functional form and mechanically increases with added regressors.

“Nonsignificant terms should always be removed.”

A term can be needed for hierarchy, confounding control, design structure, or a prespecified scientific estimand.

“Residuals should be tiny.”

Residual magnitude reflects σ^2 ; adequacy concerns structure, not simply small numerical residuals.

Practice: reason from the model space

PRACTICE

- Suppose the true mean is quadratic in x but you fit only an intercept and linear term. Express the mean residual vector in matrix notation.
- Show that adding q regressors cannot increase training SSE. What projection matrix represents the extra fitted component?
- If the larger model contains the true mean, why does adding irrelevant regressors not bias the fitted mean?
- What happens to residual degrees of freedom and average leverage as p increases?
- Under an underfit model, what is the distribution of SSE/σ^2 and what is its noncentrality parameter?
- Why can a model chosen by searching for the smallest p -values have invalid naive post-selection inference?

Takeaways

SUMMARY

Mean-space view

Correct specification means $\mu \in C(X)$.

Underfitting

μ has a component outside $C(X)$: bias and systematic residual structure remain.

Overfitting

The model contains more directions than needed: training SSE falls, but variance and leverage can rise.

Quadratic forms

Underfitting makes residual SSE noncentral; the lack-of-fit term is $\|(I-H)\mu\|^2$.

Inference

A prespecified overfit model can still have exact classical inference; data-driven selection is different.

Model choice

Use science, diagnostics, nested tests, validation, or penalized criteria according to the inferential goal.

Model complexity is a subspace choice: too small creates bias; too large spends precision.